Graph theory and extremal combinatorics

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L5-CO1 Level 5 Advanced · Combinatorics

Trees, bipartite graphs, the theorems of Mantel, Turán and Hall, and small Ramsey numbers, with the extremal arguments that prove them and the olympiad problems they solve.
By the end you can translate a problem into a graph and use degrees, trees and odd cycles to prove structural facts; prove and apply Mantel, Turán and Hall, and find the matching extremal constructions; run longest-path, shortest-odd-cycle and maximum-degree arguments to olympiad standard.
Prerequisites: Invariants, monovariants and extremal arguments (L4-CO1), Recursion, bijections and double counting (L4-CO2). **Contests:** BMO2, EGMO, USAMO, IMO. **Time:** about 180 minutes.

1 Lesson

Among ten people there are 26 pairs of friends. Prove that some three of them are mutual friends.

Suppose not. Write $d(v)$ for the number of friends of v. If x and y are friends, they have no common friend, so $d(x)+d(y)\le 10$. Add this over all 26 friendships. Each person v appears in $d(v)$ of these inequalities, so the left-hand side adds up to $\sum_v d(v)^2$, and by the QM-AM inequality

$$\sum_v d(v)^2 \;\ge\; \frac{1}{10}\Big(\sum_v d(v)\Big)^2 = \frac{52^2}{10} > 260 = 26\cdot 10.$$

This is a contradiction. The same computation shows that a triangle-free graph on n vertices has at most $n^2/4$ edges (Mantel's theorem), and $26 > 100/4$.

This unit covers the graph theory that BMO2, EGMO and USAMO problems assume. I assume the extremal principle (L4-CO1) and double counting (L4-CO2).

1.1 Language

A graph G is a finite set V of vertices with a set E of edges, each edge an unordered pair of distinct vertices. Write $n=|V|$ and $e=|E|$. The degree $d(v)$ is the number of edges at v, and $N(v)$ is the set of neighbours of v. A path is a sequence of distinct vertices with consecutive ones adjacent. A cycle is a path of at least three vertices whose ends are also adjacent, and its length is its number of edges. G is connected if any two vertices are joined by a path. The maximal connected pieces are the components. An independent set contains no edge, and a clique contains every possible edge. K_n is the complete graph, and $K_{a,b}$ the complete bipartite graph with parts of sizes a and b.

Theorem 1 (Handshake lemma).
$\sum_{v\in V} d(v) = 2e$. In particular, the number of vertices of odd degree is even.

Proof. Each edge is counted once from each of its two ends. ■

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1.2 Trees

A tree is a connected graph with no cycle. A forest is a graph with no cycle, so its components are trees. A leaf is a vertex of degree 1.

Theorem 2.
(a) Every tree with $n \geq 2$ vertices has at least two leaves. (b) A tree with n vertices has $n-1$ edges, and a forest with n vertices and c components has $n-c$ edges. (c) A connected graph with n vertices and $n-1$ edges is a tree.

Proof. (a) Take a longest path $x_0x_1 \dots x_k$ (extremal principle). Then $k \geq 1$ since there is an edge. Every neighbour of x_0 lies on the path, or the path could be extended. If x_0 were adjacent to some x_i with $i \geq 2$, then $x_0x_1 \dots x_ix_0$ would be a cycle. So $d(x_0) = 1$, and similarly $d(x_k) = 1$.

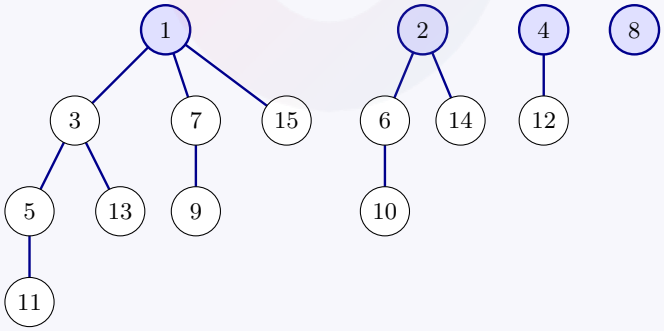
(b) Induct on n. Deleting a leaf from a tree with $n \geq 2$ vertices leaves a tree with one vertex and one edge fewer. For a forest, add the formula over the components.

(c) While the graph has a cycle, delete one edge of a cycle. This keeps the graph connected, since the rest of the cycle still joins the two ends. We finish with a tree, which by (b) has $n-1$ edges, so no edge was deleted. ■

The longest-path argument proves more. If every vertex has degree at least $d \geq 2$, then the end x_0 has at least d neighbours on the path, the furthest being some x_i with $i \geq d$, so G has a cycle of length at least $d+1$.

Example 1.
In how many ways can the integers $1, 2, \dots, 100$ be coloured red and blue so that no two different integers of the same colour add up to a power of 2?

Solution.
Make a graph on $\{1, \dots, 100\}$, joining a and b when $a+b$ is a power of 2. We want the number of proper 2-colourings of this graph. Draw this graph for $1, \dots, 15$.



The picture is a forest, and the reason is visible. Each number has at most one smaller neighbour. Indeed, if $a < b$ and $a + b = 2^k$, then $b < 2^k < 2b$, and the open interval $(b, 2b)$ contains at most one power of 2, because consecutive powers of 2 differ by a factor of 2. The interval contains none if b is a power of 2, and exactly one otherwise.
A graph in which every vertex has at most one smaller neighbour has no cycle. The largest vertex of a cycle would have two smaller neighbours on that cycle. So the graph is a forest. The numbers up to 100 with no smaller neighbour are $1, 2, 4, 8, 16, 32, 64$, so the number of edges is $100 - 7 = 93$, and by Theorem 2(b) there are $100 - 93 = 7$ components.

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A tree has exactly two proper 2-colourings. Colour any vertex, and the colour of every other vertex is forced along the unique path to that vertex, with no conflict since there is no cycle. Colourings of different components are independent, so the answer is $2^7 = 128$.

1.3 Bipartite graphs and odd cycles

G is bipartite if V splits into two independent sets, that is, if G has a proper colouring with two colours.

Theorem 3.
A graph is bipartite if and only if the graph has no cycle of odd length.

Proof. A cycle in a bipartite graph alternates between the parts, so the cycle has even length. Conversely, suppose there is no odd cycle. First, a closed walk of odd length contains an odd cycle. If no vertex repeats, the walk is a cycle. If some vertex repeats, the walk splits at that vertex into two shorter closed walks whose lengths add to an odd number, so one of them is odd, and we induct on the length. Now work in one component, fix a vertex r, and colour each vertex by the parity of its distance from r. If two adjacent vertices u, v had the same colour, then a shortest path from r to u, the edge uv and a shortest path from v back to r would form a closed walk of odd length $\operatorname{dist}(r,u) + \operatorname{dist}(r,v) + 1$, hence an odd cycle. ■

In extremal problems a *shortest* odd cycle is often the right object to take, since such a cycle has no chords. A chord would split the cycle into two shorter cycles, one of them odd.

1.4 Mantel and Turán

Theorem 4 (Mantel).
A triangle-free graph on n vertices has at most $\lfloor n^2/4 \rfloor$ edges, with equality only for $K_{\lfloor n/2 \rfloor, \lceil n/2 \rceil}$.

Proof. Let v have the largest degree Δ. Since G has no triangle, $N(v)$ is independent, so every edge has an end in $S = V \setminus N(v)$, a set of $n - \Delta$ vertices each of degree at most Δ. Hence

$$e \;\le\; \sum_{s \in S} d(s) \;\le\; (n-\Delta)\Delta \;\le\; \frac{n^2}{4}.$$

For equality every inequality must be tight. No edge has both ends in S, every $s \in S$ has degree $\Delta = |N(v)|$ and so is adjacent to all of $N(v)$, and $\Delta(n-\Delta) = \lfloor n^2/4 \rfloor$ forces $\{\Delta, n-\Delta\} = \{\lfloor n/2 \rfloor, \lceil n/2 \rceil\}$. So G is complete bipartite with parts S and $N(v)$. ■

Theorem 5 (Turán).
If a graph on n vertices contains no K_{r+1}, then $e \le \left(1 - \frac{1}{r}\right) \frac{n^2}{2}$.

Proof. Induct on r. For $r = 1$ there are no edges. Let $r \ge 2$ and let v have the largest degree Δ. The graph induced on $N(v)$ has no K_r (with v this would give a K_{r+1}), so by induction this graph has at most $\left(1 - \frac{1}{r-1}\right) \frac{\Delta^2}{2}$ edges. Every other edge has an end in $V \setminus N(v)$, so there are at most $(n-\Delta)\Delta$ of them. Therefore

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$$e \le \frac{r-2}{r-1}\cdot\frac{\Delta^2}{2} + (n-\Delta)\Delta = n\Delta - \frac{r}{2(r-1)}\Delta^2 \le \frac{r-1}{r}\cdot\frac{n^2}{2},$$

where the last step is the maximum of a quadratic in Δ, attained at $\Delta = \frac{r-1}{r}n$. ■

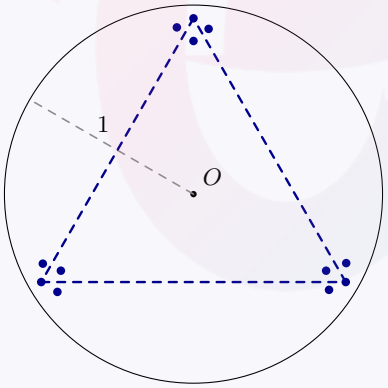
The bound is attained when $r \mid n$ by the complete r-partite graph with equal parts. (For general n the maximum is attained by the complete r-partite graph with parts as equal as possible.)

Example 2.
Twelve distinct points lie in a closed disc of radius 1. At most how many of the 66 pairs of points can be at distance greater than $\sqrt{2}$?

Solution.
Join two points when their distance exceeds $\sqrt{2}$. Since $\sqrt{2}$ is the side of a square inscribed in the circle, test four points. Can four points be pairwise more than $\sqrt{2}$ apart? If one of them is the centre O, then O is within 1 of the others. Otherwise the four rays from O through the points divide the full angle, so two of them make an angle $\theta \le 90°$. For those two points P, Q, the cosine rule gives

$$PQ^2 = OP^2 + OQ^2 - 2\,OP\cdot OQ\cos\theta \le OP^2 + OQ^2 \le 2.$$

So the graph has no K_4, and Theorem 5 with $r = 3$ gives at most $\frac{2}{3}\cdot\frac{144}{2} = 48$ edges.
The equality case of Turán's theorem tells us what to build, namely three groups of four with every pair from different groups far apart. Put four points very close to each vertex of an equilateral triangle inscribed in the circle. Points in different groups are at distance close to $\sqrt{3} > \sqrt{2}$, so $3\cdot 4\cdot 4 = 48$ pairs are far apart. The answer is 48.



1.5 Hall's theorem

A matching is a set of edges with no common ends. In a bipartite graph with parts A and B, for $S \subseteq A$ write $N(S)$ for the set of vertices of B adjacent to some vertex of S.

Theorem 6 (Hall).
A bipartite graph with parts A, B has a matching covering every vertex of A if and only if $|N(S)| \ge |S|$ for every $S \subseteq A$.

Proof. The condition is necessary, since the partners of S are $|S|$ distinct vertices of $N(S)$. For sufficiency, induct on $|A|$. The case $|A| = 1$ is clear.

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Case 1. Here $|N(S)| \geq |S| + 1$ for every non-empty $S \subsetneq A$. Match any $a \in A$ with any neighbour b and delete both. In the remaining graph each non-empty $S \subseteq A \setminus \{a\}$ has lost at most the one neighbour b, so S still has at least $|S|$ neighbours, and induction finishes.

Case 2. Some non-empty $S_0 \subsetneq A$ has $|N(S_0)| = |S_0|$. Hall's condition holds inside the graph on $S_0 \cup N(S_0)$, so by induction S_0 can be matched into $N(S_0)$. In the graph on $(A \setminus S_0) \cup (B \setminus N(S_0))$, a set $T \subseteq A \setminus S_0$ has neighbourhood $N(T \cup S_0) \setminus N(S_0)$, of size at least $|T| + |S_0| - |S_0| = |T|$. So induction matches $A \setminus S_0$ there too, and the two matchings together cover A. ■

Corollary 7.
If every vertex of a bipartite graph (multiple edges allowed) has degree $k \geq 1$, then the graph has a perfect matching.

Proof. The $k|S|$ edges leaving $S \subseteq A$ all end in $N(S)$, which receives $k|N(S)|$ edges in total, so $|N(S)| \geq |S|$. Also $k|A| = e = k|B|$, so a matching covering A is perfect. ■

Example 3.
Each of the numbers $1, 2, \ldots, 6$ is written in six cells of a 6×6 table, one number per cell. Prove that one can choose a cell in each column so that the six chosen numbers are all different.

Solution.
Choosing greedily column by column fails, since an early choice can use up the only number a later column could take. This is a matching problem. Let A be the six columns and B the six numbers, and join column c to number k if k appears in c. We need a matching covering A.
Check Hall's condition. A set S of columns contains $6|S|$ cells, and every number in them belongs to $N(S)$. Each number occupies only six cells of the whole table, so the cells of S hold at most $6|N(S)|$ entries. Hence $6|S| \leq 6|N(S)|$, and Theorem 6 gives the matching. (Equivalently, join column c to number k by one edge for each appearance of k in c. This multigraph is 6-regular, and Corollary 7 applies.)

1.6 Ramsey numbers

$R(s,t)$ is the least n such that every red/blue colouring of the edges of K_n has a red K_s or a blue K_t.

Theorem 8.
$R(3,3) = 6$, and $R(s,t) \leq R(s-1,t) + R(s,t-1)$ for $s, t \geq 2$.

Proof. Let $n = R(s-1,t) + R(s,t-1)$ and fix a vertex v of a coloured K_n. Its $n-1$ edges split into r red and b blue with $r + b = n - 1$, so $r \geq R(s-1,t)$ or $b \geq R(s,t-1)$. In the first case the red neighbours of v contain a blue K_t, or a red K_{s-1} which with v is a red K_s. The second case is symmetric. With $R(2,3) = R(3,2) = 3$ this gives $R(3,3) \leq 6$, and the pentagon with red sides and blue diagonals shows $R(3,3) > 5$. ■

The values worth remembering are $R(3,3) = 6$, $R(3,4) = 9$, $R(3,5) = 14$ and $R(4,4) = 18$. Theorem 8 gives only $R(3,4) \leq 10$, and parity improves this to 9. In a colouring of K_9 with no red triangle and no blue K_4, the red neighbours of a vertex form a blue clique, so there are at most 3 of them. Its blue neighbours contain no triangle of either colour, so there are at most 5. Hence every vertex has exactly 3 red edges, and there are $\frac{9 \cdot 3}{2}$ red edges, which is absurd. Practice problem

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2 uses the same method. That $R(4,4) > 17$ needs a specific colouring of K_{17} (Greenwood and Gleason, 1955), which I will not reproduce. Lower bounds for large k come from the counting argument in L5-CO2.

1.7 Extremal arguments on graphs

Besides a longest path and a vertex of largest degree, a useful extreme object is the subgraph left after repeatedly deleting vertices of small degree.

Example 4.
A graph has 30 vertices and 121 edges. Prove that the graph contains a copy of every tree with 5 edges.

Solution.
Since $121 > 4 \cdot 30$, we guess the general statement that *a graph with n vertices and more than $(k-1)n$ edges contains every tree with k edges.* Large average degree does not obviously help, but large *minimum* degree does, so we first find a subgraph in which every degree is large.
Step 1. Repeatedly delete a vertex whose degree in the current graph is at most $k-1$. If every vertex were deleted, all edges would be removed, at most $k-1$ with each deletion, so $e \le (k-1)n$, a contradiction. So the process stops at a non-empty subgraph H in which every degree is at least k.
Step 2. Let T be a tree with k edges, hence $k+1$ vertices. Order its vertices $t_0, t_1, \dots, t_k$ so that each t_i with $i \ge 1$ has exactly one neighbour among $t_0, \dots, t_{i-1}$ (repeatedly remove a leaf, and reverse the order of removal). Place t_0 at any vertex of H. When $t_0, \dots, t_{i-1}$ have been placed at distinct vertices, t_i must go to a neighbour of the image of its earlier neighbour t_j. That image has at least k neighbours in H, and at most $i-1 \le k-1$ of them are already used (all placed vertices except the image of t_j itself). So a free neighbour exists, and the embedding never gets stuck.
With $n = 30$ and $k = 5$ this proves the claim. The constant $k-1$ is not best possible. A conjecture of Erdős and Sós asserts that more than $\frac{k-1}{2}n$ edges suffice, and that is open in general.

Common traps.

- Assuming a graph is connected. Theorem 2(b) has $n-c$, not $n-1$, for forests. Problems about "a network" often need connectivity stated or proved.
- Proving the bound in an extremal problem and forgetting the construction, or the reverse. "Find the maximum" needs both.
- Applying Hall to the wrong side. The condition is checked for subsets of the side you want covered.
- In a longest path argument, forgetting to say why every neighbour of the end lies on the path.
- Quoting Turán with the wrong r. No K_{r+1} gives $(1-\frac{1}{r})\frac{n^2}{2}$.

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1.8 How to spot this in a contest

Whenever a problem has people and friendships, cities and roads, or objects and a symmetric relation, draw the graph and translate. The standard moves, in the order I try them, are:

1. count degrees (handshake lemma, double counting edges or paths of length 2),
2. take an extreme object, for example a longest path, a shortest odd cycle or a vertex of largest or smallest degree,
3. if the question is about two-colouring or splitting into two groups, look for odd cycles (Theorem 3),
4. if a forbidden subgraph is a triangle or a clique, compare with Mantel or Turán, and expect the extremal example to be complete multipartite,
5. if each object must be assigned its own distinct partner, set up a bipartite graph and check Hall's condition,
6. if a statement is "every colouring contains a monochromatic ...", fix a vertex and use pigeonhole on its edges, as in Theorem 8.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆⋆ A tree has a vertex of degree 10 and a vertex of degree 7. What is the smallest possible number of leaves?

Problem 2 ⋆⋆ Prove that among any seven people there are three mutual friends or four people among whom at most one pair are friends. Show that this is false for six people.

Problem 3 ⋆⋆⋆ In a graph every vertex has degree at least 3. Prove that the graph has a cycle whose length is not a prime number.

Problem 4 ⋆⋆⋆ In a 7×7 table, exactly 3 cells in each row and exactly 3 cells in each column are marked. Prove that the marked cells can be coloured red, green and blue so that each row and each column contains one marked cell of each colour.

Problem 5 ⋆⋆⋆ Among nine people there are exactly 27 pairs of friends. Prove that there are at least 27 sets of three mutual friends, and show that 27 can occur.

Problem 6 ⋆⋆⋆⋆ In a club of n members no three are mutual friends. Whenever two members are friends, the number of friends of the first plus the number of friends of the second is greater than $\frac{4n}{5}$. Prove that the members can be split into two groups so that no two members of the same group are friends. Show also that, for every n divisible by 5, the conclusion can fail if "greater than" is replaced by "at least".

3 Contest ladder

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Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **IMO 1964 Problem 4** ⋆⋆
   Compare with the proof that $R(3,3) = 6$.
2. **BMO2 2009/10 Problem 1** ⋆⋆⋆
   If two children are at distance k in the friendship graph, how far apart can they stand in the line?
3. **USAMO 2001 Problem 1** ⋆⋆⋆
   Count pairs of balls sharing a box in two ways, then construct an example meeting the bound.
4. **BMO2 2005/06 Problem 4** ⋆⋆⋆
   For each question, look at the set of children who got that question wrong, and see what the condition says about two such sets.
5. **USAJMO 2022 Problem 2** ⋆⋆⋆
   Think of a choice of cells, no two in a row or column, as a matching between rows and columns.
6. **BMO2 2013/14 Problem 1** ⋆⋆⋆
   For the lower bound, find a large family of diagonals that pairwise cross. Then look for a colouring that matches this family.
7. **IMO 1992 Problem 3** ⋆⋆⋆⋆
   Use $R(3,3) = 6$, and think about how few edges can be left uncoloured.
8. **BMO2 2004/05 Problem 4** ⋆⋆⋆⋆
   Consider how many sets the most popular element lies in, and split into cases.
9. **USAMO 2007 Problem 4** ⋆⋆⋆⋆
   Think of the animal as a connected graph on its cells and look for a good place to cut the graph.
10. **IMO 2019 Problem 3** ⋆⋆⋆⋆
    Find properties of the graph that the moves can be made to preserve, and reduce the number of vertices one at a time.

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4 Hints

1. Use the handshake lemma and $e = n - 1$ to express the number of leaves in terms of the degrees greater than 2.
2. Assume not. Bound the number of friends and the number of strangers of one person, as in the proof that $R(3,4) \leq 9$, and then look for a person with few friends.
3. Take a longest path. Its first vertex has at least three neighbours, all on the path, so you get three cycles. How are their lengths related?
4. Rows and columns are the two parts of a bipartite graph whose edges are the marked cells. What does Corollary 7 give, and what is left after removing the matching?
5. Two friends x and y have at least $d(x) + d(y) - 9$ common friends. Add this over all friendships and count what you get.
6. By Theorem 3 we only need to rule out odd cycles. Take a shortest odd cycle and ask how many neighbours a vertex can have on the cycle.

5 Solutions

Problem 1

A tree has a vertex of degree 10 *and a vertex of degree* 7*. What is the smallest possible number of leaves?*

Let the tree have n vertices, n_1 of them leaves. By Theorem 2 the tree has $n - 1$ edges, so the handshake lemma gives $\sum_v d(v) = 2n - 2$, that is, $\sum_v \big(d(v) - 2\big) = -2$. Leaves contribute -1 each, vertices of degree 2 contribute 0, and no vertex has degree 0 (a tree with a vertex of degree 10 has $n \geq 2$ and is connected). Hence

$$n_1 = 2 + \sum_{d(v) \geq 3} \big(d(v) - 2\big) \geq 2 + (10 - 2) + (7 - 2) = 15.$$

Equality is possible. Take adjacent vertices u and w, attach 9 leaves to u and 6 leaves to w. This is a tree with 17 vertices, $d(u) = 10$, $d(w) = 7$ and 15 leaves. The answer is 15.

Answer: 15

Problem 2

Prove that among any seven people there are three mutual friends or four people among whom at most one pair are friends. Show that this is false for six people.

Call a set of four people *sparse* if at most one pair among them are friends. Suppose seven people have no three mutual friends and no sparse set, and fix a person v.

The friends of v are pairwise strangers, since two friends of v who were friends would form three mutual friends with v. Four friends of v would form a sparse set, so v has at most 3 friends.

Let S be the set of strangers of v. If some $x \in S$ were a stranger to two other members y, z of S, then the five pairs vx, vy, vz, xy, xz would be strangers and $\{v, x, y, z\}$ would be sparse. So the pairs of strangers inside S are disjoint. If $|S| \geq 5$, take five members of S. At most two disjoint

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pairs of strangers lie among them, so taking one person from each such pair together with the people in no such pair, we find three of the five who are mutual friends. Hence $|S| \leq 4$, and v has at least $6 - 4 = 2$ friends. Moreover, if $|S| = 4$ then S contains exactly two disjoint pairs of strangers (with at most one, three members of S would be mutual friends), so the friendships inside S form a 4-cycle $abcd$, with a, c strangers and b, d strangers.

So every person has 2 or 3 friends. They cannot all have 3, since the number of pairs of friends would then be $\frac{7 \cdot 3}{2}$. Now fix a person v with exactly 2 friends x and y. They are strangers, and the 4 strangers of v form a 4-cycle of friendships $abcd$ as above.

Each of a, b, c, d has two friends on the cycle and at most 3 friends in all, so at most one friend in $\{x, y\}$. Suppose a has no friend in $\{x, y\}$. Then the strangers of a are v, c, x, y, so by the third paragraph, applied to a in place of v, their friendships form a 4-cycle. Since v, c are strangers and x, y are strangers, this cycle is $vxcy$, and c is friends with b, d, x and y, which is too many. Hence each of a, b, c, d has exactly one friend in $\{x, y\}$, which gives four friendships between $\{a, b, c, d\}$ and $\{x, y\}$.

Now x is a friend of v and has at most 3 friends, so x has at most 2 friends among a, b, c, d, and the same holds for y. Therefore each of x, y has exactly 2 friends among a, b, c, d. Two friends of x are not friends of each other, so they are opposite on the cycle. Say x is friends with a and c. Then in $\{v, x, b, d\}$ the only pair of friends is v, x, so this set is sparse, a contradiction.

For six people, split them into two groups of three and let two people be friends exactly when they are in different groups. Any three people include two from the same group, so no three are mutual friends, and four people split between the groups as $2 + 2$ or $3 + 1$, so they include 4 or 3 pairs of friends.

Problem 3

In a graph every vertex has degree at least 3. Prove that the graph has a cycle whose length is not a prime number.

Let $x_0 x_1 \ldots x_k$ be a longest path in the graph. Every neighbour of x_0 lies on the path, since otherwise the path could be extended at x_0. As $d(x_0) \geq 3$, the vertex x_0 has neighbours x_1, x_i and x_j with $1 < i < j \leq k$.

Consider the three cycles

$$x_0 x_1 \ldots x_i x_0, \qquad x_0 x_1 \ldots x_j x_0, \qquad x_0 x_i x_{i+1} \ldots x_j x_0,$$

which are genuine cycles because the vertices of the path are distinct, $i \geq 2$ and $j > i$. Their lengths are $p = i + 1$, $q = j + 1$ and $q - p + 2$. If p or q is not prime, we are done. Otherwise p and q are primes with $3 \leq p < q$, so both are odd. Then $q - p + 2$ is even and at least 4, so the third cycle has a length that is not prime.

(The hypothesis cannot be weakened to degree at least 2. A cycle of prime length has every degree equal to 2 and no other cycle.)

Problem 4

In a 7×7 table, exactly 3 cells in each row and exactly 3 cells in each column are marked. Prove that the marked cells can be coloured red, green and blue so that each row and each column contains one marked cell of each colour.

Form a bipartite graph whose parts are the 7 rows and the 7 columns, with one edge between row r and column c for each marked cell (r, c). Every vertex has degree 3. A colouring as required is the same as a partition of the edges into three perfect matchings, the red, green and blue cells.

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We prove by induction on k that a bipartite graph in which every vertex has degree $k \geq 1$ splits into k perfect matchings. By Corollary 7 (which follows from Hall's theorem) the graph has a perfect matching M. Removing the edges of M lowers every degree by exactly 1, because M covers every vertex exactly once. If $k = 1$ the graph is M itself. Otherwise the rest is $(k-1)$-regular and splits into $k-1$ perfect matchings by induction.

Applying this with $k = 3$ and colouring the cells of the three matchings red, green and blue, each row and each column meets each matching in exactly one edge, so every row and every column contains one marked cell of each colour.

Problem 5

Among nine people there are exactly 27 pairs of friends. Prove that there are at least 27 sets of three mutual friends, and show that 27 can occur.

Write $d(v)$ for the number of friends of v, and T for the number of sets of three mutual friends. Let x and y be friends. Among the other 7 people, x has $d(x) - 1$ friends and y has $d(y) - 1$ friends, so they have at least $(d(x) - 1) + (d(y) - 1) - 7 = d(x) + d(y) - 9$ common friends.

Each set of three mutual friends contains three friendships, and is counted once for each of them when we add up the numbers of common friends over all friendships. Hence

$$3T \;\geq \sum_{xy \text{ friends}} \big(d(x) + d(y) - 9\big) = \sum_{v} d(v)^2 - 9 \cdot 27,$$

since v appears in exactly $d(v)$ friendships. By the handshake lemma $\sum_v d(v) = 54$, so by the QM-AM inequality $\sum_v d(v)^2 \geq \frac{54^2}{9} = 324$. Therefore $3T \geq 324 - 243 = 81$ and $T \geq 27$.

For equality, split the people into three groups of three and make two people friends exactly when they are in different groups. There are $3 \cdot 3 \cdot 3 = 27$ friendships, and a set of three is a set of mutual friends exactly when the set has one person from each group, so there are 27 such sets.

(The same argument shows that any graph with n vertices and e edges has at least $\frac{e(4e-n^2)}{3n}$ triangles, which is positive as soon as $e > n^2/4$. This is a quantitative form of Mantel's theorem.)

Problem 6

In a club of n members no three are mutual friends. Whenever two members are friends, the number of friends of the first plus the number of friends of the second is greater than $\frac{4n}{5}$. Prove that the members can be split into two groups so that no two members of the same group are friends. Show also that, for every n divisible by 5, the conclusion can fail if "greater than" is replaced by "at least".

Consider the friendship graph G. By Theorem 3 we only need to show that G has no odd cycle. Suppose G has one, and let C be an odd cycle of smallest length m. Since there is no triangle, $m \geq 5$.

Claim 1. Two vertices of C are adjacent only if they are consecutive on C. A chord would split C into two cycles of lengths $a + 1$ and $b + 1$, where $a, b \geq 2$ and $a + b = m$. One of $a + 1$, $b + 1$ is odd, because their sum $m + 2$ is odd, and both are less than m. This contradicts the minimality of m. So every vertex of C has exactly 2 neighbours on C.

Claim 2. A vertex v not on C has at most 2 neighbours on C. Suppose v has three neighbours on C. They cut C into three arcs of lengths $\ell_1, \ell_2, \ell_3 \geq 1$ with $\ell_1 + \ell_2 + \ell_3 = m$, and each arc together with v forms a cycle of length $\ell_i + 2$. Since m is odd, either exactly one ℓ_i is odd or all three are. If

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exactly one, say ℓ_1, is odd, then $\ell_2, \ell_3 \geq 2$, so $\ell_1 \leq m-4$ and the cycle of length $\ell_1 + 2 \leq m-2$ is odd and shorter than C. If all three are odd, take the smallest, $\ell_1 \leq \frac{m}{3}$. Then $\ell_1 + 2 \leq \frac{m}{3} + 2 < m$ because $m \geq 5$, and again we have a shorter odd cycle. Both cases contradict the minimality of m.

Now count the edges between C and V in two ways. Every vertex of G, on C or not, has at most 2 neighbours on C, so

$$\sum_{c \in C} d(c) = \sum_{v \in V} |N(v) \cap C| \;\leq\; 2n.$$

On the other hand, add the hypothesis over the m edges of C. Each vertex of C is an end of exactly two of these edges, so

$$2 \sum_{c \in C} d(c) = \sum_{xy \text{ an edge of } C} \big(d(x) + d(y)\big) \;>\; m \cdot \frac{4n}{5} \;\geq\; 4n,$$

since $m \geq 5$. This contradicts the first inequality, so G has no odd cycle and is bipartite, which is the required split.

For $n = 5k$, put the members into five groups $B_1, \ldots, B_5$ of k each, and let two members be friends exactly when their groups are B_i and B_{i+1} for some i, where $B_6 = B_1$. Each member has $2k$ friends, so any two friends have $4k = \frac{4n}{5}$ friends together. Three mutual friends would lie in three different groups, each consecutive to the other two in the cyclic order $B_1, \ldots, B_5$, which is impossible. But one member from each group gives a 5-cycle, so by Theorem 3 the required split does not exist.

(This strengthens a theorem of Andrásfai, Erdős and Sós, whose hypothesis is that every member has more than $\frac{2n}{5}$ friends. That hypothesis implies the one here.)